Activity based

Project Report on

Computer Network

Submitted to Vishwakarma University, Pune

Under the Initiative of

Contemporary Curriculum, Pedagogy, and Practice (C2P2)

By

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Project Report : 01

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Design and Implementation of a Local Area Network (LAN) for a Specific Location

Project Statement :

Design and implement a Local Area Network (LAN) for a specified location, such as a school, office building, or community center. The project involves identifying optimal spots for network devices (switches, routers, and access points), configuring the network with appropriate IP addresses, establishing communication between different sub-networks, and calculating the overall cost of network development. This project aims to provide students with hands-on experience in network design, configuration, and cost analysis.

Project Objective: To understand development stages in Local Area Network

Project Outcome: Student can develop Local Area Network and perform cost analysis of Network Infrastructure.

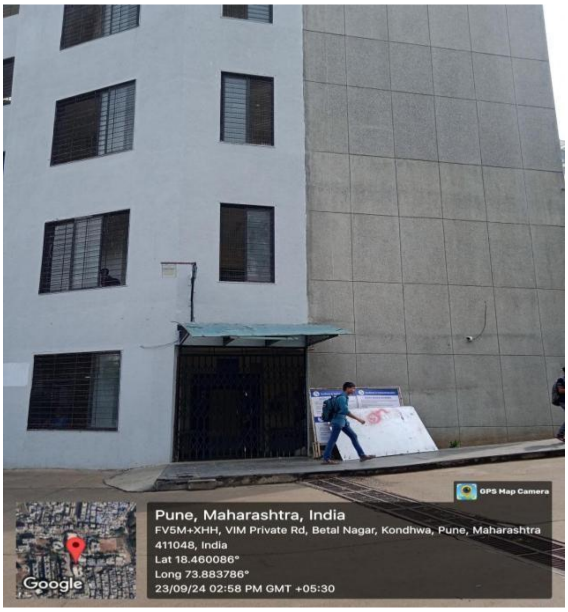
Theory:

Site Survey:

Fig no:1

Fig no:2

Fig no:3



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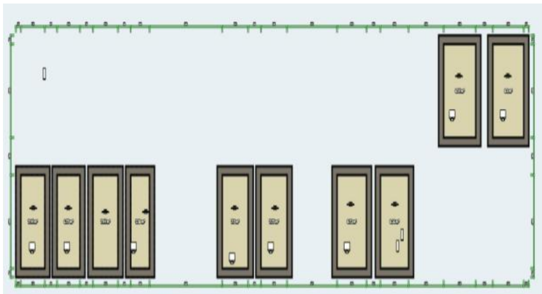

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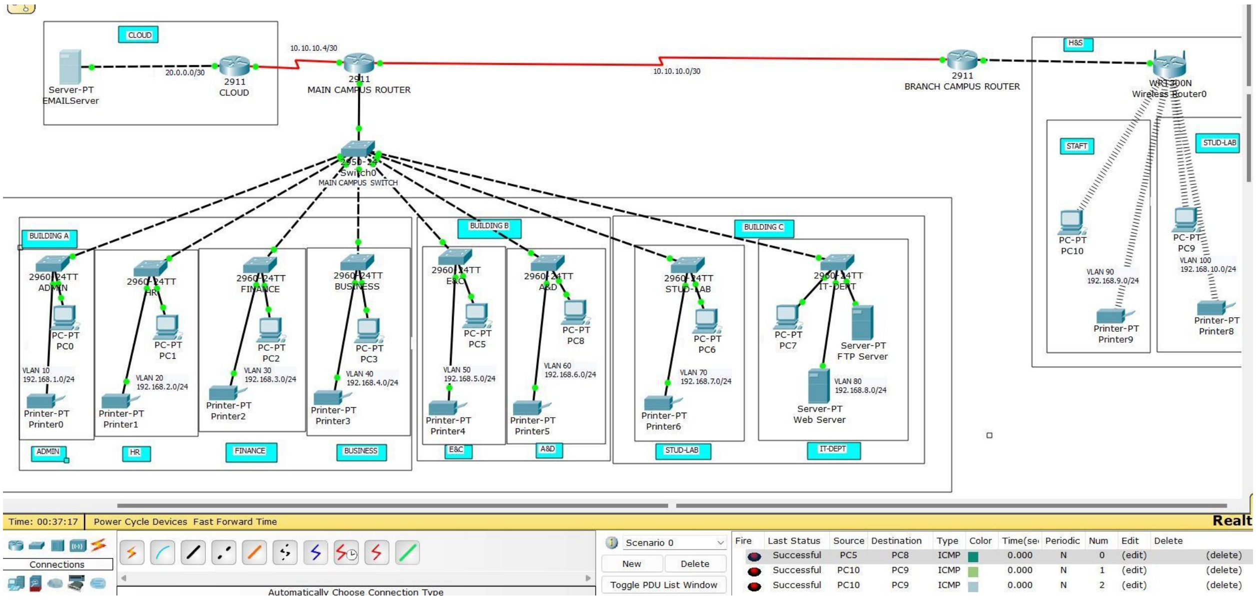
Network Design Implementation:



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1. Network Topology:

- **Type:** Hierarchical Star.
- **Core Router:** Cisco 2911 connects the main campus to the internet and branch campus.
- **Distribution Layer:** Cisco 2950 switch links all buildings and handles VLAN routing.
- **Access Layer:** Each department has a Cisco 2960 switch for device connectivity.

2. Device Placement:

- **Core Router (Main Campus):** Centrally located Cisco 2911 for external communications and VLAN routing.
- **Main Campus Switch:** Cisco 2950 handles inter-VLAN routing.
- **Department Switches:** Cisco 2960 per department (e.g., Admin, Exam, Finance).
- **Branch Campus:** Cisco 2911 router and 3560 switch.
- **Servers:** Email (cloud), FTP and web servers (IT dept).

3. IP Addressing:

- **Private IPs:** 192.168.0.0 range.
- **VLANs:** Separate VLANs for each department (e.g., Admin: 192.168.1.0/24, HR: 192.168.2.0/24).
- **Router Links:** 20.0.0.0/30 (cloud) and 10.10.10.0/30 (branch).

4. VLANs:

- VLANs configured on switches with inter-VLAN routing via Layer 3 switches/routers.

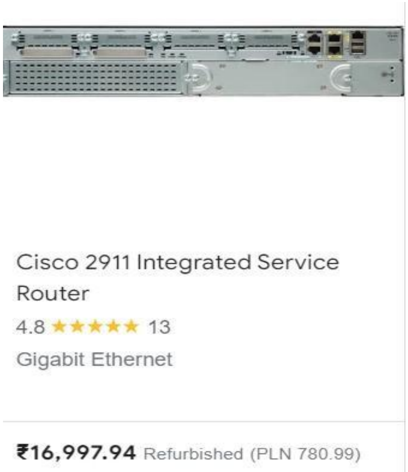
5. Security & Redundancy:

- **STP:** Prevents loops.
- **Port Security & ACLs:** Limit device access and control traffic.
- **DHCP Snooping & ARP Inspection:** Enhances network security.

6. Network Services:

- **DHCP:** Auto-assigns IPs.
- **DNS:** Internal resolution via web server.
- **NTP:** Syncs time on all devices.
- **Logging:** Centralized logging via syslog server.

Cost Analysis Implementation:



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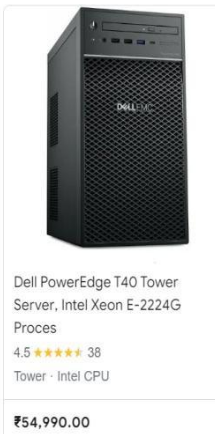

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Item	Quantiy	Real-Time Estimated Cost
Cisco 2911 Router	3	50,991
Cisco 2960 Switch	9	74,340
WR-300N Wireless Router	1	899
Servers (FTP, Email, Web)	3	1,64,970
Desktop Computers (PCs)	10	1,54,990
Printers	9	77,391
Installation Labor		16,000

Final Total Cost: 5,39,581 lakhs

Conclusion :

The design and implement a Local Area Network (LAN) for a real-world scenario. I have learned how to plan the placement of routers, switches, & other network devices, and how to configure them to ensure proper communication between different departments and campuses. The project taught me about VLANs, IP addressing, and how to manage network traffic securely and efficiently.The key takeaways for me was understanding how to balance functionality and cost. By choosing more affordable hardware and using open-source software, I realized that it's possible to build a reliable network without exceeding the budget.